Gondwana reconstruction, finite rotation poles, models:

Plate Model CR26BAAE (Hotspot Reference Frame)

Africa versus hotspot reference frame

400	10.000	-40.880	139.310	1.950	
400	20.000	-40.880	139.310	3.900	
400	30.000	-40.880	139.310	5.850	
400	40.000	-40.880	139.310	7.800	
400	46.540	-40.880	139.310	9.000	
400	72.500	-35.597	144.888	14.524	
400	100.500	-28.461	145.638	21.691	
400	130.000	-23.340	145.152	30.294	
400	200.000	-14.554	146.585	41.021	
400	270.000	-12.754	142.409	54.561	
400	300.000	-16.161	135.487	61.424	
400	340.000	-17.010	121.112	76.978	**CR26BAAE**

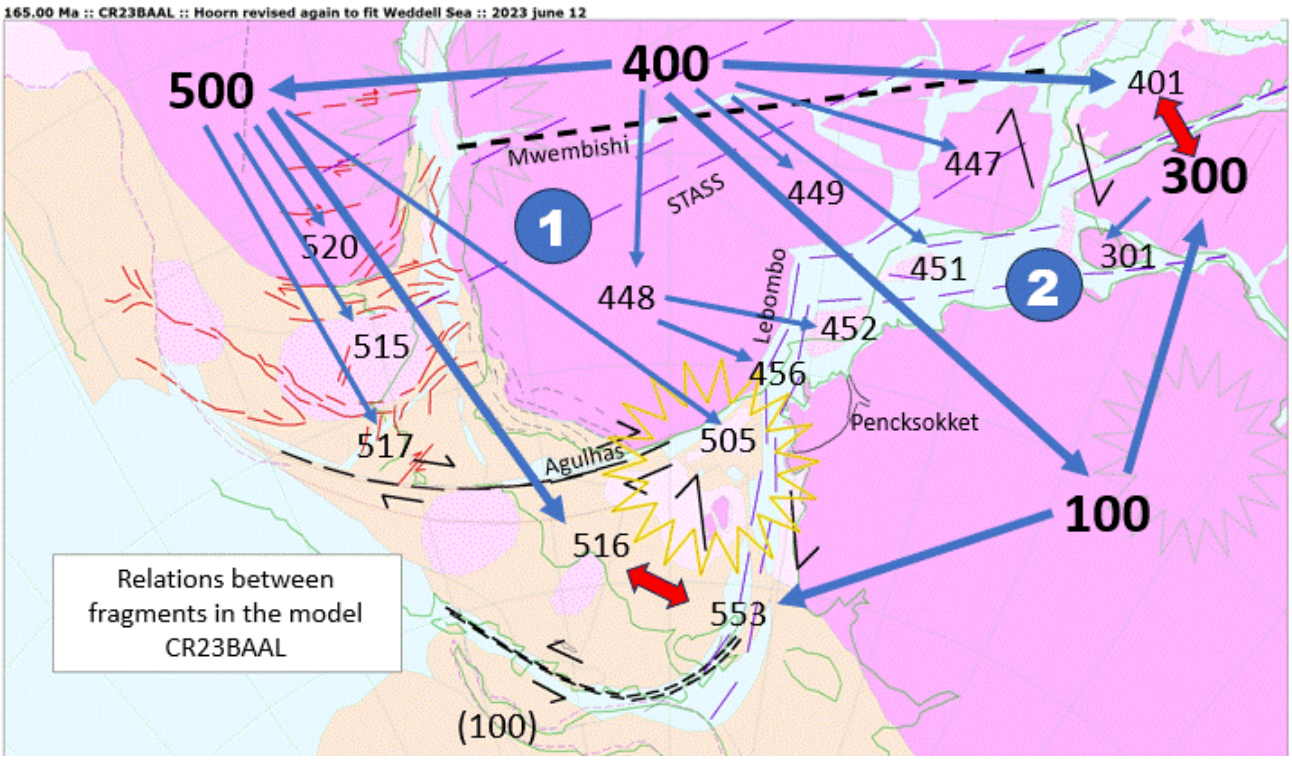

Figure 1. **The structure of the plate model in two principal plate circuits, West Gondwana (1) and East Gondwana (2), that share the relation of Antarctica to Africa. The closure of each circuit is indicated in red. The motion of Africa with respect to a global reference frame is defined in the table at the head of this page. In all the subsequent tables, the motion of the fragments is presented with respect to a fixed Africa.**

Plate model CR26AAAE (Africa fixed)

Plate Circuit 1: 553-100-448-400-500-516 (West Gondwana)

100 Antarctica versus Africa

100	41.000	7.820	-38.920	7.800	
100	43.960	8.446	-42.921	8.267	
100	51.000	-1.566	-40.775	9.761	
100	67.000	-8.494	-37.885	12.701	
100	70.900	-9.763	-37.147	14.202	
100	72.500	-72.967	-66.129	11.230	
100	80.000	-68.615	-55.438	12.153	
100	80.000	-7.932	-34.407	18.097	
100	83.640	-7.752	-34.141	18.597	**C34**
100	100.500	-0.855	-34.388	27.092	Base Cenomanian
100	113.200	-3.057	-31.771	34.527	Base Albian
100	121.400	-7.823	-28.238	40.629	**M0**, base Aptian
100	124.700	-11.112	-25.367	44.040	**M3/M2**
100	127.500	-12.313	-24.205	46.275	**M5/M4**
100	130.680	-11.904	-24.678	47.691	**M10**
100	134.700	-11.385	-25.398	48.789	**M11**
100	137.700	-10.916	-26.044	49.828	**M14**
100	142.300	-10.381	-26.776	51.072	**M18**
100	154.940	-9.009	-28.633	54.575	base Kimmeridgian
100	184.200	-9.799	-31.297	57.885	base Toarcian
100	230.000	-9.799	-31.297	57.885	
100	300.000	-10.885	-31.591	58.034	**CR26AAAE**

448 Sub-Palala-Zoetfontein Africa versus Africa

455	121.400	90.000	0.000	0.000	
455	154.940	57.340	-15.340	0.500	
455	555.000	57.340	-15.340	0.500	**CR26AAAE**

123 Mid-ocean ridge in Weddell Sea versus Africa

123	32.720	84.155	-148.160	11.269
123	41.000	82.149	-69.605	12.663
123	43.960	79.592	-73.902	13.288
123	48.300	75.456	-58.253	13.181
123	51.000	72.946	-53.746	13.171
123	58.200	68.677	-47.295	13.890
123	65.000	66.736	-45.616	14.588
123	67.000	66.442	-44.714	15.014
123	70.900	63.242	-42.998	16.047
123	80.000	57.534	-39.319	19.941
123	83.000	58.173	-39.092	20.787
123	83.640	55.490	-37.502	20.485
123	99.000	51.056	-42.131	28.977

123	100.500	51.446	-42.667	29.916	
123	113.200	49.949	-42.754	36.200	
123	121.400	36.330	-35.267	37.551	
123	124.700	30.444	-31.395	38.853	
123	127.500	27.534	-29.904	40.170	
123	127.520	27.530	-29.908	40.179	
123	130.680	27.405	-29.889	41.618	
123	134.700	27.282	-30.665	42.720	
123	137.700	29.283	-31.472	43.876	
123	140.490	27.960	-31.206	44.674	
123	142.300	27.817	-31.287	45.207	
123	147.000	27.308	-32.545	46.372	
123	154.940	26.907	-33.914	48.732	
123	165.000	26.419	-34.510	49.767	
123	184.200	24.505	-36.947	50.966	
123	230.000	24.505	-36.947	50.966	
123	300.000	23.377	-37.166	50.470	**CR26AAAE**

500 South America versus Africa

500	22.000	61.850	-40.740	8.350	
500	38.000	62.892	-36.920	15.088	
500	43.960	63.093	-36.100	18.086	
500	48.000	63.219	-35.898	19.084	
500	53.000	63.435	-35.546	21.082	
500	67.000	63.727	-35.060	24.569	
500	71.900	63.899	-34.768	27.217	
500	83.640	61.603	-35.014	33.403	
500	100.500	58.858	-37.471	44.325	
500	113.200	54.636	-35.264	50.160	
500	124.700	50.589	-32.540	53.103	
500	142.300	46.807	-30.584	55.908	
500	555.000	46.807	-30.584	55.908	**CR26AAAE**

516 Malvinas Plateau (Hoorn) versus Africa

516	22.000	61.850	-40.740	8.350	
516	38.000	62.892	-36.920	15.088	
516	43.960	63.093	-36.100	18.086	
516	48.000	63.219	-35.898	19.084	
516	53.000	63.435	-35.546	21.082	
516	67.000	63.727	-35.060	24.569	
516	71.900	63.899	-34.768	27.217	
516	83.640	61.603	-35.014	33.403	
516	100.500	58.858	-37.471	44.325	
516	113.200	54.636	-35.264	50.160	
516	121.400	45.320	-32.941	52.609	
516	124.700	41.720	-31.405	54.287	
516	130.680	36.151	-29.470	56.991	
516	134.700	33.887	-28.577	58.600	
516	137.700	31.720	-27.789	60.181	**CR26AAAE**

516	142.300	29.736	-27.033	61.958	
516	154.940	24.766	-25.943	64.741	
516	165.000	19.058	-24.757	69.002	
516	555.000	19.058	-24.757	69.002	**CR26AAAE**

This plate circuit closes between the (conjectural) southern-and-eastern margin of the Malvinas Plateau (Hoorn, 516) and the Weddell Sea mid-ocean ridge (123). The objective has been to create a Weddell Sea mid-ocean ridge system that develops symmetrically about the ridge (123) initiated at 142.3 Ma. The elements of the plate circuit have been adjusted so as to minimise concertina-like growth of the ocean between Antarctica and 123. The relative movement of SAM and ANT now (CR26AAE) also reproduce the smooth arcs of the 'Antarctica-South America corridor' recorded in the Weddell Sea, after about 100 Ma.

Other fragments that make up the model in the area of the Bouvet triple junction are:

505 Maurice Ewing Bank versus Africa

505	22.000	61.850	-40.740	8.350	
505	38.000	62.892	-36.920	15.088	
505	43.960	63.093	-36.100	18.086	
505	48.000	63.219	-35.898	19.084	
505	53.000	63.435	-35.546	21.082	
505	67.000	63.727	-35.060	24.569	
505	71.900	63.899	-34.768	27.217	
505	83.640	61.603	-35.014	33.403	
505	100.500	58.858	-37.471	44.325	
505	113.200	54.636	-35.264	50.160	
505	121.400	53.487	-35.810	53.054	
505	124.700	50.249	-34.254	53.735	
505	130.680	48.480	-33.697	56.059	
505	142.300	42.924	-31.282	57.486	
505	154.940	39.737	-30.354	58.894	
505	555.000	39.737	-30.354	58.894	**CR26AAAE**

451 Beira High versus Africa

451	30.000	-23.330	13.870	0.150	
451	121.400	-23.330	13.870	0.150	
451	130.680	33.684	-1.865	0.348	
451	160.000	33.684	-1.865	0.348	
451	184.200	-24.410	32.413	18.135	
451	230.000	-24.410	32.413	18.135	
451	300.000	-25.086	33.092	18.050	**CR26AAAE**

452 Limpopia versus Africa

452	30.000	-23.330	13.870	0.150	
452	121.400	-23.330	13.870	0.150	
452	127.500	38.674	-132.011	1.707	
452	134.700	50.249	-113.636	2.746	
452	137.500	44.276	-92.690	3.217	
452	142.300	31.022	-70.219	4.149	
452	154.940	17.809	-72.371	8.280	
452	178.000	3.481	-70.566	10.609	
452	179.000	2.642	-69.525	10.761	
452	184.200	0.347	-69.299	11.360	
452	300.000	-1.316	-73.790	11.253	**CR26AAAE**

456 St Lucia fragment versus Africa

456	30.000	-23.330	13.870	0.150	
456	121.400	-23.330	13.870	0.150	
456	124.700	9.100	5.410	0.192	
456	130.680	-11.826	9.609	4.230	
456	134.700	-13.065	9.906	6.497	
456	178.000	-13.065	9.906	6.497	
456	179.000	-13.091	9.836	6.697	
456	230.000	-13.091	9.836	6.697	
456	300.000	-14.708	11.085	6.508	**CR26AAAE**

449 (Zimbabwe) versus Africa

449	30.000	-23.330	13.870	0.150	
449	121.400	-23.330	13.870	0.150	
449	142.300	33.684	-1.865	0.348	
449	230.000	33.684	-1.865	0.348	
449	300.000	32.099	67.640	0.124	**CR26AAAE**

447 (North Mozambique) versus Africa

447	5.000	-23.330	13.870	0.300	
447	121.400	-23.330	13.870	0.300	
447	130.680	10.964	6.118	0.491	
447	142.300	16.564	4.832	0.575	
447	230.000	16.564	4.832	0.575	
447	300.000	-22.655	41.397	3.006	**CR26AAAE**

There are also three fragments between South America (500) and the Hoorn fragment (516):

520 Uruguay versus Africa

520	22.000	61.850	-40.740	8.350
520	38.000	62.892	-36.920	15.088

520	43.960	63.093	-36.100	18.086	
520	48.000	63.219	-35.898	19.084	
520	53.000	63.435	-35.546	21.082	
520	67.000	63.727	-35.060	24.569	
520	71.900	63.899	-34.768	27.217	
520	83.640	61.603	-35.014	33.403	
520	100.500	58.858	-37.471	44.325	
520	113.200	54.636	-35.264	50.160	
520	124.700	50.589	-32.540	53.103	
520	130.680	49.267	-31.821	54.030	
520	142.300	46.151	-30.201	56.513	**CR26AAAE**

515 South of BA No1 versus Africa

515	22.000	61.850	-40.740	8.350	
515	38.000	62.892	-36.920	15.088	
515	43.960	63.093	-36.100	18.086	
515	48.000	63.219	-35.898	19.084	
515	53.000	63.435	-35.546	21.082	
515	67.000	63.727	-35.060	24.569	
515	71.900	63.899	-34.768	27.217	
515	83.640	61.603	-35.014	33.403	
515	100.500	58.858	-37.471	44.325	
515	113.200	54.636	-35.264	50.160	
515	124.700	50.589	-32.540	53.103	
515	128.000	49.855	-32.135	53.611	
515	142.300	41.103	-29.204	58.239	**CR26AAAE**

517 South of BA No2 versus Africa

517	22.000	61.850	-40.740	8.350	
517	38.000	62.892	-36.920	15.088	
517	43.960	63.093	-36.100	18.086	
517	48.000	63.219	-35.898	19.084	
517	53.000	63.435	-35.546	21.082	
517	67.000	63.727	-35.060	24.569	
517	71.900	63.899	-34.768	27.217	
517	83.640	61.603	-35.014	33.403	
517	100.500	58.858	-37.471	44.325	
517	113.200	54.636	-35.264	50.160	
517	124.700	44.367	-32.028	53.431	
517	128.000	41.945	-31.523	54.200	
517	142.300	33.804	-28.529	59.373	**CR26AAAE**

Plate Circuit 2: 401-400-448-100-300 (East Gondwana)

100 Antarctica versus Africa

100	41.000	7.820	-38.920	7.800	
100	43.960	8.446	-42.921	8.267	
100	51.000	-1.566	-40.775	9.761	
100	67.000	-8.494	-37.885	12.701	
100	70.900	-9.763	-37.147	14.202	
100	72.500	-72.967	-66.129	11.230	
100	80.000	-68.615	-55.438	12.153	
100	80.000	-7.932	-34.407	18.097	
100	83.640	-7.752	-34.141	18.597	**C34**
100	100.500	-0.855	-34.388	27.092	Base Cenomanian
100	113.200	-3.057	-31.771	34.527	Base Albian
100	121.400	-7.823	-28.238	40.629	**M0**, base Aptian
100	124.700	-11.112	-25.367	44.040	**M3/M2**
100	127.500	-12.313	-24.205	46.275	**M5/M4**
100	130.680	-11.904	-24.678	47.691	**M10**
100	134.700	-11.385	-25.398	48.789	**M11**
100	137.700	-10.916	-26.044	49.828	**M14**
100	142.300	-10.381	-26.776	51.072	**M18**
100	154.940	-9.009	-28.633	54.575	base Kimmeridgian
100	184.200	-9.799	-31.297	57.885	base Toarcian
100	230.000	-9.799	-31.297	57.885	
100	300.000	-10.885	-31.591	58.034	**CR26AAAE**

401 Madagascar versus Africa

401	20.000	-23.330	13.870	0.300	
401	117.300	-23.330	13.870	0.300	
401	121.400	5.897	-94.991	2.131	
401	124.700	7.510	-98.917	5.122	
401	127.500	7.933	-99.963	8.169	
401	130.680	8.083	-100.337	10.368	
401	142.300	4.436	-94.382	14.154	
401	154.940	1.866	-86.618	17.531	
401	184.200	-5.453	-83.612	20.811	
401	230.000	-5.453	-83.612	20.811	
401	300.000	-8.483	-83.831	20.702	**CR26AAAE**

India versus Africa

300	33.430	-13.342	-125.663	19.395
300	41.000	-14.843	-131.474	24.759
300	42.860	-14.871	-132.610	26.450
300	43.960	-14.746	-133.866	27.278
300	51.000	-18.005	-141.081	31.514
300	67.000	-19.898	-153.217	41.909

300	70.900	-20.816	-154.773	44.244	
300	72.500	-21.007	-155.227	45.038	
300	80.000	-22.203	-156.431	49.618	
300	83.640	-22.073	-157.783	52.499	
300	89.000	-21.387	-157.281	57.220	
300	100.500	-21.364	-157.194	57.265	
300	113.200	-21.324	-156.990	57.513	
300	117.300	-21.430	-157.056	57.261	
300	121.400	-21.579	-154.734	58.540	
300	124.700	-21.800	-152.071	59.949	
300	127.500	-22.121	-149.748	61.204	
300	130.680	-22.259	-147.631	62.669	
300	134.700	-22.682	-146.415	63.244	
300	137.700	-23.127	-145.305	63.705	
300	142.300	-23.576	-143.989	64.424	
300	154.940	-25.633	-140.517	64.844	
300	184.200	-28.689	-138.528	66.566	**CR26AAAE**

448 Southern Kalahari versus Africa

448	30.000	-23.330	13.870	0.150	
448	121.400	-23.330	13.870	0.150	
448	130.680	33.684	-1.865	0.348	
448	178.000	33.684	-1.865	0.348	
448	179.000	16.685	1.882	0.504	
448	230.000	16.685	1.882	0.504	
448	300.000	3.704	27.730	0.263	**CR26AAAE**

Fragments 448 (Southern Kalahari), *449 (Zimbabwe*) and ***447 (North Mozambique)*** are listed under Plate Circuit 1.

This plate circuit closes between India and Madagascar. The movements of Madagascar against Africa and of India against Antarctica have been adjusted to minimise/eliminate relative movement between Madagascar and India before 89 Ma. The movement of India against Antarctica has been refined to ensure steady strike-slip movement of (Greater)India along (a) the long transform off Western Australia, 142.3 to 100.5 Ma and (b) the Davie Fracture Zone 134.7 to 89 Ma, i.e. while Madagascar is still fully attached to India.

200 Australia versus Africa

200	33.430	-15.002	-126.879	19.523
200	41.000	-15.523	-125.401	22.265
200	42.860	-15.321	-125.286	23.277
200	43.960	-14.746	-124.796	23.556
200	51.000	-16.414	-121.175	24.217
200	67.000	-15.686	-114.756	25.647
200	70.900	-16.076	-111.725	26.224
200	80.000	-14.809	-103.277	27.495

200	83.640	-13.901	-102.393	27.903	
200	100.500	-10.322	-94.091	34.988	
200	113.200	-14.412	-83.324	37.969	
200	121.400	-20.709	-74.865	40.208	
200	124.700	-25.181	-69.729	41.194	
200	127.500	-27.036	-66.709	42.258	
200	130.680	-26.712	-65.589	43.547	
200	134.700	-26.172	-65.054	44.739	
200	137.700	-25.688	-64.579	45.864	
200	142.300	-25.139	-64.049	47.206	
200	154.940	-23.753	-62.736	50.961	
200	184.200	-24.500	-63.045	55.100	**CR26AAAE**

Australia does not form part of either plate circuit. Rotations are included for completeness. The Australia-Antarctica poles are taken largely from published work supplemented with a closer fit to Antarctica in conformity with the principles adopted throughout our Gondwana reassembly.

301 Sri Lanka versus Africa

301	33.430	-13.342	-125.663	19.395	
301	41.000	-14.843	-131.474	24.759	
301	42.860	-14.871	-132.610	26.450	
301	43.960	-14.746	-133.866	27.278	
301	51.000	-18.005	-141.081	31.514	
301	67.000	-19.898	-153.217	41.909	
301	70.900	-20.816	-154.773	44.244	
301	72.500	-21.007	-155.227	45.038	
301	80.000	-22.203	-156.431	49.618	
301	83.640	-22.073	-157.783	52.499	
301	89.000	-21.387	-157.281	57.220	
301	100.500	-21.364	-157.194	57.265	
301	109.000	-21.337	-157.057	57.431	
301	112.200	-16.957	-149.648	63.343	
301	113.200	-15.779	-147.929	65.364	
301	117.300	-11.558	-141.963	73.885	
301	121.400	-8.278	-136.220	85.360	
301	124.700	-10.418	-136.459	82.764	
301	127.500	-12.144	-136.560	80.248	
301	130.680	-12.829	-135.637	80.842	
301	134.700	-13.878	-135.247	80.062	
301	137.700	-14.789	-134.787	79.543	
301	142.300	-15.986	-134.309	78.662	
301	154.940	-17.899	-131.792	79.683	
301	184.200	-20.598	-130.708	81.605	
301	230.000	-20.598	-130.708	81.605	
301	300.000	-21.183	-131.343	81.748	**CR26AAAE**

Sri Lanka is confined between India and Antarctica, escaping first by growth of ocean between it and India, then between it and Antarctica.

404 Madagascar Rise versus Africa

404	20.000	-23.330	13.870	0.300	
404	63.500	-23.330	13.870	0.300	
404	72.500	7.604	-43.820	2.112	
404	89.000	10.350	-48.603	7.098	
404	117.300	10.350	-48.603	7.098	
404	121.400	11.438	-60.671	8.656	
404	124.700	11.976	-71.000	11.108	
404	127.500	12.122	-77.581	13.823	
404	130.680	12.136	-80.908	15.857	
404	142.300	8.965	-80.763	19.798	
404	154.940	6.332	-77.190	23.496	
404	184.200	0.202	-76.404	26.730	
404	230.000	0.202	-76.404	26.730	
404	300.000	-2.134	-76.625	26.525	**CR26AAAE**

CVR

Delft, 2026 April 22

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